

Data Session 5 Key, part 1: Following the Money

84-355 Democracy's Data | Session 5 — Money, message, and attention | Instructor copy
— do not distribute

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All Part 8 code is given to students. Grade the interpretation.

Every number below was run against the data file.

The lab in one sentence

Organised money in American politics is not distributed by competitiveness, ideology, or merit — it is distributed by **who currently holds the seat** — and the ledger that proves this is also incapable of telling you who won.

The three findings

```
h <- fec[fecc$office == "House", ]
hs <- h[h$ici %in% c("I", "C", "O"), ]

c(median = round(median(h$ttr_receipts)), mean = round(mean(h$ttr_receipts)))
```

```
#> median    mean
#> 22583 782177
```

```
aggregate(ttr_receipts ~ ici, data = hs, FUN = median)
```

```
#>   ici ttr_receipts
#> 1   C    9106.745
#> 2   I 1537104.825
#> 3   O   19272.080
```

```
ser <- hs[hs$ttl_receipts > 50000, ]
ser$pac_share <- 100 * ser$pac_contrib / ser$ttl_receipts
aggregate(pac_share ~ ici, data = ser, FUN = function(x) round(median(x), 1))
```

```
#>   ici pac_share
#> 1   C         1.1
#> 2   I        37.6
#> 3   0         1.4
```

1. **Concentration.** Median House candidate raises **\$22,583**; mean is **\$782,177**, $\sim 35\times$ higher. Top 10% of candidates raise **72.7%** of all House money; bottom 50% raise **0.2%**.
2. **Incumbency.** Median incumbent **\$1,537,105**; median challenger **\$9,107**. A ratio of **169:1**.
3. **PAC money follows office, not contests.** Among serious House campaigns ($> \$50k$), PACs supply **37.6%** of the median incumbent's money, **1.1%** of a challenger's, **1.4%** in an open seat.

Finding 3 is the one to spend class time on. Finding 2 is ambiguous between “money makes incumbents” and “incumbents attract money.” Finding 3 is not: an open-seat race is *maximally* competitive and PACs fund it at 1.4%. They are not buying influence over outcomes; they are buying access to people who already have power.

Option A — Does the Senate work the same way?

Verified output

```
o <- fec[fec$office == "Senate" & fec$ici %in% c("I","C","O") &
        fec$ttl_receipts > 50000, ]
o$pac_share <- 100 * o$pac_contrib / o$ttl_receipts
aggregate(pac_share ~ ici, data = o, FUN = function(x) round(median(x), 1))
```

```
#>   ici pac_share
#> 1   C         0.2
#> 2   I        15.0
#> 3   0         0.7
```

```
aggregate(ttl_receipts ~ ici, data = o, FUN = function(x) round(median(x)))
```

```
#>   ici ttl_receipts
#> 1   C       574755
#> 2   I      2845748
#> 3   0       947895
```

```
table(o$ici)
```

```
#>
#>   C   I   0
#> 80 102  66
```

Senate PAC shares: incumbents **15.0%**, challengers **0.2%**, open seats **0.7%**. Same direction, much smaller magnitude. Sample is small — 102 incumbents, 80 challengers, 66 open-seat candidates above the threshold.

What a good answer says

3/4: Reports that the pattern holds but is weaker, and notices the small sample.

4/4: Gets the mechanism. PAC contribution limits are *fixed per committee per election* while Senate campaigns are enormous. The same \$10,000 that is a meaningful slice of the \$2.0 million a typical serious House incumbent raises is a rounding error in Sherrod Brown's \$96.5 million. PAC money has not gone away in Senate races; it has been **diluted by individual money** flowing in at a scale House races never see. A student who reasons from "the denominator changed" rather than "PACs care less about senators" has understood it.

4/4 bonus: notes that this makes the House the place where PAC money has the most *leverage*, which is the opposite of where most public attention goes.

Common stumble: "PACs prefer the House." The absolute dollars say otherwise; it is the share that falls.

Option B — Where does the money come from?

Verified output

```
s <- fec[fec$office == "House" & fec$ici %in% c("I","C","O") &
      fec$ttl_receipts > 50000, ]
src <- sapply(c("C","O","I"), function(k) {
  x <- s[s$ici == k, ]
  c(Individuals = sum(x$indiv_contrib), PACs = sum(x$pac_contrib),
    Party = sum(x$party_contrib), Self = sum(x$self_funding)) /
    sum(x$ttl_receipts) * 100
})
colnames(src) <- c("Challenger", "Open seat", "Incumbent")
round(src, 1)
```

```
#>           Challenger Open seat Incumbent
#> Individuals      68.1      62.7      57.9
#> PACs             7.6       7.5      24.5
#> Party            0.2       0.1       0.1
#> Self             17.4      25.8       4.8
```

source	Challenger	Open seat	Incumbent
Individuals	68.1	62.7	57.9
PACs	7.6	7.5	24.5
Party	0.2	0.1	0.1
Self	17.4	25.8	4.8

What a good answer says

3/4: Notices the aggregate PAC gap (7.6 vs 24.5) is much narrower than the median gap (1.1 vs 37.6), and can explain why: the aggregate is dominated by a handful of enormous challenger campaigns that *do* attract PAC money, while the typical challenger gets none at all.

4/4: Identifies **self-funding as what fills the gap** — 17.4% for challengers and 25.8% in open seats, against 4.8% for incumbents. Then draws the conclusion that matters: if you want to challenge a sitting member of Congress and you are not independently wealthy, the organised money will not come, and the shortfall is made up out of the candidate’s own pocket or not at all. That is a barrier to entry with a dollar figure attached.

4/4, best: answers the “which summary is fairer” question honestly — neither; they answer different questions. The median describes the typical candidate’s experience, the aggregate describes where the money in the system actually sits. A journalist wanting a dramatic number takes 1.1% vs 37.6%. A journalist wanting an accurate picture of the flow of funds takes both.

Common stumble: treating the two sets of numbers as a contradiction, or assuming one must be an error. Address this directly — it is the most transferable statistical idea in the lab.

Option C — How big is the double-counting problem?

Verified output

```
for (fl in c(1000000, 100000, 10000)) {  
  b <- fec$ttl_receipts > fl, ]  
  d <- table(b$ttl_receipts); sh <- as.numeric(names(d[d > 1]))  
  cat("floor $", format(fl, big.mark = ","), ": ", nrow(b), " candidates, ",  
      sum(b$ttl_receipts %in% sh), " rows sharing a total\n", sep = "")  
}
```

```
#> floor $1,000,000: 745 candidates, 70 rows sharing a total  
#> floor $100,000: 1421 candidates, 130 rows sharing a total  
#> floor $10,000: 2178 candidates, 207 rows sharing a total
```

At the \$1,000,000 floor the chunk finds **35** shared totals, not a handful, and nearly all of them are genuine. Biden/Harris carry **\$1,175,189,365** on two records. **26** of the groups pair a House filing with a Senate one — members who moved committee money across when they ran for the Senate in 2024, of whom **Gallego, Trone, Schiff** and **Rosen** are only the four largest — and **Hovde** appears under two Senate IDs in the same race.

Only **2** groups at this floor put two different surnames on the same total, and one of those is Biden/Harris. The other is worth knowing before a student finds it: **Goodlander** (NH-02) and the write-in presidential candidate **Fishback** share \$4,358,978 — and, on inspection, *every other financial column in the row as well*. That is a defect in the source file rather than a coincidence, and it is the sharpest thing in this option.

What a good answer says

3/4: Reports that lowering the floor finds more apparent duplicates, and suspects some are coincidences.

4/4: Reasons about *why* coincidences appear — at low dollar values, round numbers cluster. Two candidates who each raised exactly \$12,500, or exactly \$15,000, are not the same person; both of those collisions are in the file at the \$10,000 floor. The method is most trustworthy where the amounts are large and precise to the cent, which is where it was used in Part 6.

Full credit, and give it readily: a student who finds the Goodlander/Fishback pair and concludes that *size and precision are not sufficient either* has beaten the key's own argument with the data. Do not mark that wrong.

4/4, best: answers the “what would you actually need” question. The right key is not the dollar amount but the **candidate's committee ID or their name plus state** — and better still, the FEC's own linkage between candidate IDs and committee IDs, which exists in a different bulk file. A student who says “I would join on identity, not on a value that happens to be equal” has learned Session 1's lesson properly.

Option D — Their own question

Reward curiosity. Debt, cash-on-hand, and single-state comparisons all work.

Grading

Four-point scale from the syllabus.

- **Never penalise code.** It is your code.
- **The central intellectual move** is distinguishing “money causes winning” from “winning attracts money.” Part 4 is designed so that the open-seat column settles it. Students who use that column in their argument have done the real work; students who only compare incumbents to challengers have not.
- **Reward students who notice a number is doing less than it appears to.** The median-versus-aggregate split in Option B is the sharpest instance.

Anticipated questions

- “*Doesn't this prove money buys elections?*” — No, and the lab is careful not to claim it. It shows money *follows* office. Whether it also causes winning is the question the FEC file cannot answer, which is Part 7.
- “*Where is the super PAC money?*” — Not here. This file covers candidates' own committees only. Independent expenditures are reported separately and are enormous post-*Citizens United*. Worth saying out loud, because students will otherwise conclude they have seen all the money. They have seen perhaps half.
- “*Why does the FEC not record who won?*” — Because it regulates finance, not elections. This is the same structural fact as Session 1: no national election administrator exists. Connect the two explicitly.
- “*Is \$9,107 really the median challenger?*” — Yes. Most challengers are not serious candidates and raise almost nothing. This is itself a finding about how few congressional races are genuinely contested — and it points straight back at Session 4's **13 competitive states**.

Connections forward

- **Session 4 (returns over time)** — the vanishing battleground and the 169:1 funding gap are the same phenomenon measured two ways. Uncontested seats do not attract money.

- **campaign-visits** (retired, on disk) — money and attention both concentrate on the same small set of races. Ask whether the causation runs in a circle.
- **media-attention, the demo in this same session** — most of this money is spent on advertising, which is where the media unit picks the thread up.
- **Session 1 (election returns)** — the FEC file is authoritative and *still* needs deduplication and still lacks outcomes. Authority is not the same as completeness.
- The assigned readings this week (the Trump donor op-ed and the Harris donors interactive) both describe the **individual** side of the ledger. Part 4 is the organised side. Ask on Tuesday which one the coverage over-weights, and why.